

HIRUSHA FERNANDO

Full-stack Software Engineer | Blockchain & Web3 Development

Sri Lanka | hirushafernando121@gmail.com | +94 726390027

[LinkedIn Profile](#) | [Github Profile](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer with close to 2 years of production experience designing and architecting scalable enterprise systems, including multi-tenant ERP and FinTech platforms. Experienced in building secure financial workflows, backend architectures, and automation solutions using Laravel, PHP, and AWS. Applying full-stack engineering expertise to Web3 engineering with hands-on experience designing, developing, and deploying Solidity smart contracts for decentralized escrow systems, collateralized lending protocols, and blockchain-based payment solutions. Strong understanding of smart contract security principles including reentrancy protection, access control, and checks-effects-interactions (CEI), with practical experience in Foundry, Hardhat, and OpenZeppelin-based development.

PROFESSIONAL EXPERIENCE

PHP Developer

NegoIT (PVT) Ltd

Mar 2026 – Present

- Leading end-to-end development of a high-traffic ERP/POS system using complex PHP and MySQL architectures.
- Designed secure financial workflows and transactional backend systems handling complex business logic, data integrity, and high-volume operations.
- Leading technical teams and mentoring developers while managing large-scale legacy systems with modern optimizations.

Software Engineer (Junior)

Appleholidays Destination Services (PVT) Ltd

Mar 2025 – Feb 2026

- Architected and maintained a multi-tenant ERP system for accounting, invoicing, and financial tracking using Laravel, React, and MySQL, integrating and consolidating data across 3 companies while supporting daily invoice processing and business operations.
- Contributed to the development and maintenance of admin and supplier dashboards, enabling efficient customer order management, supplier coordination, and real-time business operations.
- Implemented secure RESTful APIs within the admin dashboard for managing commercial hotel information and lifestyle data across multiple countries, incorporating authentication, authorization, and data validation mechanisms to ensure secure data handling.
- Assisted in AWS-based deployments, monitoring system stability, security, and uptime in production environments.

Software Engineer (Internship)

Appleholidays Destination Services (PVT) Ltd

Oct 2024 – Mar 2025

- Contributed to admin and supplier dashboard development using Laravel and React, implementing new features and collaborating with senior developers.
- Developed responsive web features using Laravel and React for internal business systems.
- Assisted in database optimization for MySQL and MongoDB, improving query efficiency and response times.
- Supported the setup of CI/CD pipelines using AWS and Kubernetes, gaining hands-on exposure to automated deployments and DevOps workflows.

EDUCATION

Bachelor of Information And Technology

Feb 2020 – Sep 2024

University of Colombo School of Computing, Sri Lanka

CORE COMPETENCIES & SKILLS

- **Smart Contracts & Web3:** Solidity, Hardhat, Foundry, OpenZeppelin, Ethers.js, Web3.js, Smart Contract Security (Reentrancy Guards, Access Control, Checks-Effects-Interactions), Chainlink Oracles, MetaMask Integration, Slither, Ethereum (Sepolia Testnet)
- **Backend & Core Languages:** Solidity, PHP, JavaScript, SQL (MySQL, PostgreSQL), HTML5, CSS3
- **Frontend & Frameworks:** Laravel, React.js, Next.js
- **Databases & Data Management:** MySQL, PostgreSQL, MongoDB (familiar with Redis, Firebase)
- **DevOps & Cloud:** AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, GitLab CI), Hostinger
- **Version Control & Collaboration:** Git, GitHub
- **Software Engineering Concepts:** OOP, Data Structures & Algorithms, RESTful API, Smart Contract Design Patterns

PROJECTS & DEVELOPMENT

Decentralized Escrow Protocol With Dispute Resolution [link](#)

Technologies: Solidity, Foundry, Hardhat, OpenZeppelin, React, Ethers.js

- Architected a milestone-based escrow smart contract system enabling secure freelance payments with built-in arbitration, eliminating the need for a trusted third party.
- Implemented industry-standard security patterns including checks-effects-interactions, reentrancy guards, and custom error handling, achieved 95%+ test coverage across smart contract state transitions, edge cases, and access-control paths using Foundry.
- Deployed and verified the contract on Ethereum Sepolia Testnet with a React/Ethers.js frontend for wallet-connected contract interaction.

Decentralized Lending Protocol [link](#)

Technologies: Solidity, Hardhat, Chainlink Oracles, OpenZeppelin, React, Ethers.js

- Engineered a collateralized lending protocol supporting WETH deposits, USDC borrowing, real-time health-factor monitoring, and automated liquidations modeled on Aave/Compound mechanics.
- Integrated Chainlink price feeds with staleness validation to ensure reliable on-chain collateral valuation for borrowing and liquidation logic.
- Built a full React dashboard for deposit, borrow, repay, and liquidation flows, deployed and live on Ethereum Sepolia Testnet.

Cross-Border Payment & Escrow System [link](#)

Technologies: Solidity, Hardhat, Chainlink Oracles, OpenZeppelin, React, Ethers.js

- Developed a USDC-based escrow and payment-processing system for cross-border settlement, supporting multi-currency payout routing (LKR, EUR, GBP, USD) via an on-chain price oracle.
- Designed a full-stack architecture with L1 escrow and L2 payment-processor contracts, backed by a Next.js frontend with wallet integration and transaction history tracking.
- Wrote unit and integration tests covering escrow creation, fee calculation, and refund logic using Hardhat.

LANGUAGES

- English – Professional Proficiency
- Sinhala – Native Proficiency